

1. LOCKOUT / TAGOUT PROCEDURE

Before any service work, the authorized technician must follow the company-mandated lockout/tagout (LOTO) procedure:

Step 1. Notify all affected operators that the machine is being serviced.

Step 2. Identify all energy sources: electrical (main disconnect), pneumatic (air supply valve), hydraulic (pressure bleed), mechanical (flywheel/spring potential energy).

Step 3. Shut down the machine using the normal stop procedure.

Step 4. Disconnect or isolate each energy source. Lock the disconnect switch in the OFF position with a personal padlock.

Step 5. Dissipate stored energy: bleed hydraulic pressure, discharge capacitors, lower suspended loads, release spring tension.

Step 6. Verify zero energy state by attempting to start the machine.

Step 7. Apply a tag identifying the technician and the expected completion time.

WARNING: Never attempt to service a machine without completing the above steps. Failure to lock out can result in severe injury or death.

2. CROSS-MACHINE ERROR CODE NOTES

E101 ? This code has different meanings on different machines in this facility:

- RoboInject-300: winding overtemperature (see RoboInject manual §4.2)
- Press-2000: low hydraulic pressure (see Press-2000 manual §3.1)
- Press-2001: drive chain binding (see Press-2001 manual §7.1)

Always identify the machine model before diagnosing any error code. Applying the wrong machine's corrective action can cause further damage and create safety hazards.

E204 ? Also varies by machine:

- RoboInject-300: servo stall (§8.3)
- Press-2000: high pressure exceeded (§5.2)
- Press-2001: flywheel overspeed (§7.2)

Common codes (same across all):

E011 ? Safety loop open

E099 ? Emergency stop activated

3. GENERAL SAFETY RULES

3.1 PPE Requirements

All personnel in the production area must wear hearing protection (earplugs or earmuffs rated ?25 dB NRR), safety glasses with side shields, and steel-toed boots. Gloves are prohibited around rotating machinery.

3.2 Housekeeping

Keep the area around each machine clear of debris, oil spills, and unused tools. Any spill exceeding 0.5 L must be reported to the shift supervisor.

3.3 Emergency Procedures

In the event of an injury, press the nearest E-stop and call for assistance. First-aid kits are located at each bay entrance. Fire extinguishers (Type ABC) are mounted at 20m intervals along the pillars.

3.4 Reporting Near Misses

Any near-miss incident (e.g., a tool dropped inside a machine guard, a hydraulic line that burst while the machine was idle) must be reported to safety within 24 hours using form SAF-101.